



NAME OF THE EXPERIMENT: DATABASE
MANAGEMENT SYSEMS

COURSE CODE: CSE2007

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SLOT : L35+L36&L47+L48

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Assignment-1

1.creating employee table

```
create table EMPLOYEE (  
  
    fname varchar(50),  
  
    minit varchar(1),  
  
    lname varchar(50),  
  
    ssn varchar(10) primary key,  
  
    bdate date,  
  
    address varchar(100),  
  
    sex varchar(1),  
  
    salary number,  
  
    super_ssn varchar(10),  
  
    dno number  
);
```

Inserting records in employee table:

```
INSERT INTO EMPLOYEE
```

```
(Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)
```

```
VALUES
```

```
('Franklin', 'T', 'Wong', '333445555', '1955-12-08',
```

```
'638 Voss, Houston, TX', 'M', 40000, '888665555', 5);
```

INSERT INTO EMPLOYEE

VALUES

***('Alicia', 'J', 'Zelaya', '999887777', '1968-01-19',
'3321 Castle, Spring, TX', 'F', 25000, '987654321', 4);***

INSERT INTO EMPLOYEE

VALUES

***('Jennifer', 'S', 'Wallace', '987654321', '1941-06-20',
'291 Berry, Bellaire, TX', 'F', 43000, '888665555', 4);***

INSERT INTO EMPLOYEE

VALUES

***('Ramesh', 'K', 'Narayan', '666884444', '1962-09-15',
'975 Fire Oak, Humble, TX', 'M', 38000, '333445555', 5);***

INSERT INTO EMPLOYEE

VALUES

***('Joyce', 'A', 'English', '453453453', '1972-07-31',
'5631 Rice, Houston, TX', 'F', 25000, '333445555', 5);***

INSERT INTO EMPLOYEE

VALUES

('Ahmad', 'V', 'Jabbar', '987987987', '1969-03-29',

'980Dallas,Houston,TX', 'M', 25000, '987654321', 4);

INSERT INTO EMPLOYEE

VALUES

('James', 'E', 'Borg', '888665555', '1937-11-10',

'450 Stone, Houston, TX', 'M', 55000, NULL, 1);

Displaying the table:

*SELECT * FROM EMPLOYEE;*

FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPER_SSN	DNO
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble,TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000		1

2.creating department table:

*CREATE TABLE DEPARTMENT (Dname VARCHAR(50), Dnumber NUMBER
PRIMARY KEY, Mgr_ssn VARCHAR(10), Mgr_start_date DATE);*

Inserting the records:

*INSERT INTO DEPARTMENT (dname, dnumber, mgr_ssn, mgr_start_date) VALUES
('Research', 5, '333445555', '1988-05-22');*

*INSERT INTO DEPARTMENT (dname, dnumber, mgr_ssn, mgr_start_date) VALUES
('Administration', 4, '987654321', '1995-01-01');*

INSERT INTO DEPARTMENT (dname, dnumber, mgr_ssn, mgr_start_date) VALUES ('Headquarters', 1, '888665555', '1981-06-19');

Display the table:

*SELECT * FROM DEPARTMENT;*

DNAME	DNUMBER	MGR_SSN	MGR_START_DATE
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

3. Creating the project table:

CREATE TABLE PROJECT (pname VARCHAR(50), pnumber NUMBER PRIMARY KEY, plocation VARCHAR(50), dnum NUMBER);

Inserting the records:

INSERT INTO PROJECT (pname, pnumber, plocation, dnum) VALUES ('ProductX', 1, 'Bellaire', 5);

INSERT INTO PROJECT (pname, pnumber, plocation, dnum) VALUES ('ProductY', 2, 'Sugarland', 5);

INSERT INTO PROJECT (pname, pnumber, plocation, dnum) VALUES ('ProductZ', 3, 'Houston', 5);

INSERT INTO PROJECT (pname, pnumber, plocation, dnum) VALUES ('Computerization', 10, 'Stafford', 4);

INSERT INTO PROJECT (pname, pnumber, plocation, dnum) VALUES ('Reorganization', 20, 'Houston', 1);

INSERT INTO PROJECT (pname, pnumber, plocation, dnum) VALUES ('Newbenefits', 30, 'Stafford', 4);

Display the table:

```
SELECT * FROM PROJECT;
```

PNAME	PNUMBER	PLOCATION	DNUM
-----	-----	-----	----
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

4.Creating the works on table:

```
CREATE TABLE WORKS_ON (essn VARCHAR(10), pno NUMBER, hours NUMBER,  
PRIMARY KEY (essn, pno));
```

Inserting the records:

```
INSERT INTO WORKS_ON (essn, pno, hours) VALUES ('123456789', 1, 32.5); INSERT  
INTO WORKS_ON (essn, pno, hours) VALUES ('123456789', 2, 7.5); INSERT INTO  
WORKS_ON (essn, pno, hours) VALUES ('666884444', 3, 40.0); INSERT INTO WORKS_ON  
(essn, pno, hours) VALUES ('453453453', 1, 20.0); INSERT INTO WORKS_ON (essn, pno,  
hours) VALUES ('453453453', 2, 20.0); INSERT INTO WORKS_ON (essn, pno, hours)  
VALUES ('333445555', 2, 10.0); INSERT INTO WORKS_ON (essn, pno, hours) VALUES  
( '333445555', 3, 10.0); INSERT INTO WORKS_ON (essn, pno, hours) VALUES ('333445555',  
10, 10.0); INSERT INTO WORKS_ON (essn, pno, hours) VALUES ('333445555', 20, 10.0);  
INSERT INTO WORKS_ON (essn, pno, hours) VALUES ('999887777', 30, 30.0); INSERT  
INTO WORKS_ON (essn, pno, hours) VALUES ('999887777', 10, 10.0); INSERT INTO  
WORKS_ON (essn, pno, hours) VALUES ('987987987', 10, 35.0); INSERT INTO  
WORKS_ON (essn, pno, hours) VALUES ('987987987', 30, 5.0); INSERT INTO WORKS_ON  
(essn, pno, hours) VALUES ('987654321', 30, 20.0);
```

```
INSERT INTO WORKS_ON (essn, pno, hours) VALUES ('987654321', 20, 15.0);
INSERT INTO WORKS_ON (essn, pno, hours) VALUES ('888665555', 20, NULL);
```

Display the table:

```
SELECT * FROM WORKS_ON;
```

ESSN	PNO	HOURS
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	

5. Creating DEPT_LOCATIONS table:

```
CREATE TABLE DEPT_LOCATIONS (dnumber NUMBER, dlocation VARCHAR(50),
PRIMARY KEY (dnumber, dlocation));
```

Inserting records:

```
INSERT INTO DEPT_LOCATIONS (dnumber, dlocation) VALUES (1, 'Houston'); INSERT
INTO DEPT_LOCATIONS (dnumber, dlocation) VALUES (4, 'Stafford'); INSERT INTO
DEPT_LOCATIONS (dnumber, dlocation) VALUES (5, 'Bellaire');
```

```
INSERTINTODEPT_LOCATIONS (dnumber, dlocation) VALUES (5, 'Sugarland');
INSERTINTODEPT_LOCATIONS (dnumber, dlocation) VALUES (5, 'Houston');
```

Display the table:

```
SELECT * FROM DEPT_LOCATIONS;
```

DNUMBER	DLOCATION
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

6. Creating *DEPENDENT TABLE*:

```
CREATE TABLE DEPENDENT (essn VARCHAR(10), dependent_name VARCHAR(50),
sex VARCHAR(1), bdate DATE, relationship VARCHAR(50));
```

Inserting records:

```
INSERT INTO DEPENDENT (essn, dependent_name, sex, bdate, relationship) VALUES
('333445555', 'Alice', 'F', '1986-04-05', 'Daughter');
```

```
INSERT INTO DEPENDENT (essn, dependent_name, sex, bdate, relationship) VALUES
('333445555', 'Theodore', 'M', '1983-10-25', 'Son');
```

```
INSERT INTO DEPENDENT (essn, dependent_name, sex, bdate, relationship) VALUES
('333445555', 'Joy', 'F', '1958-05-03', 'Spouse');
```

```
INSERT INTO DEPENDENT (essn, dependent_name, sex, bdate, relationship) VALUES
('987654321', 'Abner', 'M', '1942-02-28', 'Spouse');
```

```
INSERT INTO DEPENDENT (essn, dependent_name, sex, bdate, relationship) VALUES
('123456789', 'Michael', 'M', '1988-01-04', 'Son');
```

```
INSERTINTODEPENDENT (essn, dependent_name, sex, bdate, relationship) VALUES
('123456789','Alice','F','1988-12-30', 'Daughter');
```

```
INSERTINTODEPENDENT (essn, dependent_name, sex, bdate, relationship) VALUES
('123456789','Elizabeth','F', '1967-05-05', 'Spouse');
```

Display the table:

```
SELECT * FROM DEPENDENT;
```

ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Questions:

1.Fetch all records of all employees having ‘s’ in their name.

```
SELECT * FROM EMPLOYEE
WHERE fname LIKE '%s%' OR lname LIKE '%s%';
```

OUTPUT:

FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPER_SSN	DNO
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000		1

2.Find the average, minimum, maximum salary from employee table.

```
SELECT AVG(salary) AS average_salary,
```

```
MIN(salary)ASminimum_salary,  
  
MAX(salary)ASmaximum_salary  
  
FROM EMPLOYEE;
```

Output:

AVERAGE_SALARY	MINIMUM_SALARY	MAXIMUM_SALARY
-----	-----	-----
36571.43	25000	55000

3.Find the cumulative salary for all male and female employees separately.

```
SELECT sex, SUM(salary) AS total_salary  
  
FROM EMPLOYEE  
  
GROUP BY sex;
```

Output:

SEX	TOTAL_SALARY
---	-----
F	93000
M	158000

4.Find the Count of employees working for each department.

```
SELECT dno, COUNT(*) AS employee_count  
  
FROM EMPLOYEE  
  
GROUP BY dno;
```

Output:

DNO	EMPLOYEE_COUNT
1	1
4	3
5	3

5. Find the Total project hours for each project

```
SELECT pno, SUM(hours) AS total_hours
```

```
FROM WORKS_ON
```

```
GROUP BY pno;
```

Output:

PNO	TOTAL_HOURS
1	52.5
2	37.5
3	50
10	55
20	25
30	55

6. How many male and female dependents are there in this organization?

```
SELECT sex, COUNT(*) AS dependent_count
```

```
FROM DEPENDENT
```

```
GROUP BY sex;
```

Output:

SEX	DEPENDENT_COUNT
F	4
M	3

7.List project names running under department 5 and 1.

```
SELECT pname, dnum  
FROM PROJECT  
WHERE dnum IN (5,1);
```

Output:

PNAME	DNUM
-----	----
ProductX	5
ProductY	5
ProductZ	5
Reorganization	1

8.List the Departments spread over multiple locations.

```
SELECT dnumber  
FROM DEPT_LOCATIONS  
GROUP BY dnumber  
HAVING COUNT(dlocation) > 1;
```

Output:

DNUMBER

5